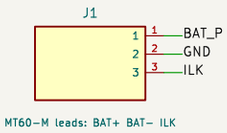
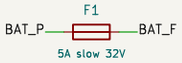
# PF233-PM2: PM2 battery dock: fuse, P-FET power switch with soft start, bulk, cord

How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.
Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

## 1 Battery input

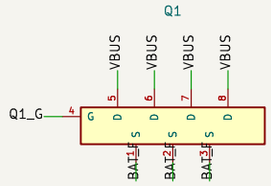
J1: MT60 leads BAT+ / BAT- / ILK (interlock), potted at the pedestal.
F1 5 A slow-blow in the BAT+ path -> BAT_F.




## 2 High-side P-FET switch with soft start

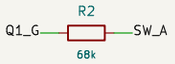
Q1 (P-channel, source on BAT_F, drain to VBUS). R1 gate-to-source 47 k; DZ1 15 V zener clamps VGS.
Gate is pulled low through R2 and the top-housing switch J3 and the ILK interlock; R3 + C1 (Miller soft start) slow the turn-on.

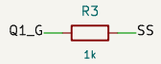


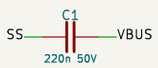


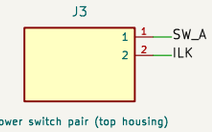




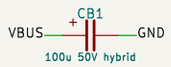



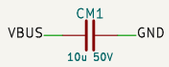



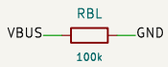

## 3 Bulk, TVS and bleeder

DTVS (30 V stand-off TVS) on VBUS; CB1 100 uF hybrid + CM1 10 uF X7R; RBL 100 k bleeder.



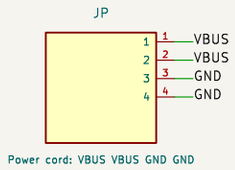



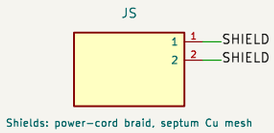




## 4 Power cord and shield

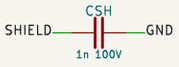
JP: coiled cord VBUS x2 / GND x2 to the head (shaft).
JS: braid and septum mesh shields; RSH + CSH drain the shield to GND at one point.



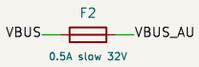


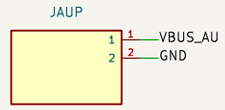





## Other parts

parts not assigned to a function above




Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
PM2 battery dock: fuse, P-FET power switch with soft start, bulk, cord
**ARX / PHI-FOLD 233**

Sheet: /
File: PF233-PM2_functional.kicad_sch

***Title: PF233-PM2 functional schematic***

| Size: User | Date: 2026-10-26 | **Rev: r3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/1 |